3.091 Aid Sheet

Atomic structure · Periodic trends · Ionic & covalent bonding — built from recitation & problem-set archetypes, L01–L09

CONSTANTS

$h = 6.626\times10^{-34}$ J·s = 4.136×10^{-15} eV·s
$\hbar = h/2\pi = 1.055\times10^{-34}$ J·s
$c = 2.998\times10^{8}$ m/s
$e = 1.602\times10^{-19}$ C | **1 eV = 1.602×10^{-19} J**
$m_e = 9.109\times10^{-31}$kg $m_p = 1.673\times10^{-27}$kg
$m_n = 1.675\times10^{-27}$kg
$N_a = 6.022\times10^{23}$ /mol
$R = 8.314$ J/mol·K = 0.08206 L·atm/mol·K
a_0 (Bohr radius) = 0.529 Å
$R_\infty = 13.6$ eV = 2.18×10^{-18} J = 1.097×10^{7} m^{-1}
$\varepsilon_0 = 8.854\times10^{-12}$ F/m

LIGHT & BOHR MODEL

$$c = \nu\lambda \quad E = h\nu = hc/\lambda$$

$$E_n = -13.6\ \text{eV}\cdot Z^2/n^2 \quad \text{(H-like ions)}$$

$$1/\lambda = R\cdot Z^2\left|1/n_c^2 - 1/n_i^2\right|$$

$$mvr = n(h/2\pi) \quad r_n = n^2 a_0$$

Series (final n): Lyman=1(UV) Balmer=2(vis) Paschen=3 Brackett=4 Pfund=5 Humphreys=6
Trap: |λ|>0 always — emission(n↓)=+ΔE sign as written; absorption flip order or take abs value.

QUANTUM NUMBERS

n	1,2,3… shell/energy
l	0…n−1 → s,p,d,f
m_l	−l…+l, orientation
m_s	±1/2 only, spin

s=1 orbital/2e⁻ · p=3/6 · d=5/10 · f=7/14
Check order: n≥1 int → 0≤l≤n−1 → $|m_l| \le l$ → $m_s = \pm½$ exactly.

PHOTOELECTRON SPECTROSCOPY

$$E_{binding} = E_{incident} - E_{photoelectron}$$

Peak **position** = binding energy → identity/Z. Peak **height/area** = e⁻ count in that orbital → configuration.
Trap: isoelectronic species (same e⁻ count) share the exact peak *pattern* but shift to different *positions* (more protons → higher BE, shifts left). 1st IE(X^+) = 2nd IE(X).

FILLING RULES

Pauli: no 2 e⁻ share all 4 QNs (max 2/orbital, opposite spin)
Aufbau: lowest energy first
Hund: fill degenerate orbitals singly (∥ spin) before pairing
n+l rule: ascending (n+l); tie → ascending n (4s n+l=4 fills before 3d n+l=5)
Exceptions (half/full-d stability): Cr=[Ar]$4s^1 3d^5$ · Cu=[Ar]$4s^1 3d^{10}$ · same pattern: Mo, Ag, Au
Trap: derive config from PT position, don't memorize. Ionizing transition metals → remove *ns* before (n−1)d, always.

PERIODIC TRENDS

Trend	→ period	↓ group
Atomic radius	↓	↑
1st IE	↑	↓
Electron affinity	↑	↓
Electronegativity	↑	↓

Driven by: nuclear charge (→) vs. shielding/distance (↓). Cation<parent atom<anion, always.
IE zig-zag (real, not typos): Be>B, Mg>Al (full s^2 easier to keep than lone p^1) · N>O, P>S (half-filled p^3 extra-stable)

IONIC BOND ENERGETICS

1. Ionize metal: +IE 2. Attach e⁻ to nonmetal: −EA 3. Coulomb attraction of ions: large −ΔE (this is what actually pays for the bond)
Trap: steps 1+2 alone are usually unfavorable — step 3 always wins. Pure element can never bond ionically with itself (needs Δχ>0).

BORN–HABER CYCLE

$$\Delta H_f = V + IE + ½B(X_2) + EA + U_L$$

2+/2− case (e.g. CaO): use IE_1+IE_2 and EA_1+EA_2.
Traps: ½B only if ½ mol diatomic gas needed/formula unit. 2nd EA step ($X^- + e^- \to X^{2-}$) is net endothermic — still used as given data.

LATTICE ENERGY (BORN–LANDÉ)

$$E = -(N_a M Z^+ Z^- e^2)/(4\pi\varepsilon_0 r_0) \times (1-1/n)$$

Kapustinskii shortcut: n=9 fixed, M≈0.88×(ion count). $U_L \propto Z^+Z^-$ (↑charge) and $\propto 1/r_0$ (↓size) — tracks melting point directly.

Structure	r^+/r^-	CN	M
Zinc blende	.225–.414	4	1.638
Rock salt	.414–.732	6	1.748
CsCl	.732–1.0	8	1.763
Fluorite (1:2)	.732–1.0	8/4	2.52

Born exp. n: He^5 Ne^7 Ar/Cu^{+9} Kr/Ag^{+10} Xe/Au^{+12} (avg of ion pair)

COVALENT: BOND TYPE

Δχ	Bond
< 0.4	pure covalent
0.4–1.8	polar covalent
> 1.8	ionic

H_2 bond curve: r_0=74pm, well depth=−432kJ/mol = bond dissociation energy. Same curve shape for ionic bonds.

LEWIS STRUCTURES — ALGORITHM

1. Arrange: least-χ atom central, H always terminal
2. Count valence e⁻ (+charge→subtract, −charge→add)
3. Odd total → radical, incomplete octet, stop
4. Draw single-bond skeleton
5. Lone pairs to terminal atoms' octets first
6. Leftover e⁻ → central atom (hypervalent if >8; only period≥3)
7. Still missing octet? → lone pair becomes double/triple bond; recheck

$$FC = \text{valence } e^- - \text{lone-pair } e^- - \text{bonding } e^-/2$$

Best structure: FC→0 (fewest nonzero); any − FC on more-electronegative atom.
FC≠oxidation state (FC splits bond 50/50; ox. state gives both e⁻ to more-χ atom).
Sanity check: N→3 bonds/1 LP · O→2 bonds/2 LP · halogen→1 bond/3 LP.
Resonance = multiple equal structures, true form is the hybrid.

DON'T FORGET — RECITATION-FLAGGED TRAPS

- Cubing volume conversions: 1 $m^3 = 10^6$ cm^3, not 10^2.
- Isotope "A≈atomic mass" shortcut only valid for single-isotope elements.
- Weighted-avg atomic mass = Σ(isotope mass × fractional abundance).
- Rydberg eq.: take |value| for λ; emission/absorption only differ by sign.
- Match units inside one equation (R_∞ in J vs eV, matching h's units).
- Config from PT position, not memorized — that's the actual tested skill.
- Transition-metal ions: always lose *ns* electrons before (n−1)d.
- 1st IE(X^+) = 2nd IE(X) — a disguised identity in comparison questions.
- Co^{2+} and friends: favor half/full d-subshell, not "isoelectronic w/ noble gas."
- PES: height=count, position=identity; isoelectronic ⇒ same pattern, shifted position.
- Be>B and N>O ionization dips are expected exceptions, not errors.
- Born–Haber: remember the ½B(X_2) factor whenever only half a mole is needed.
- O^{2-}'s 2nd electron-affinity step is net endothermic, yet still used as given data.
- Formal charge (50/50 split) ≠ oxidation state (all e⁻ to more-χ atom) — don't conflate.
- H is always terminal in a Lewis structure, never central.
- Expanded octets only exist for period ≥3 (accessible d-orbitals) — never C,N,O,F.